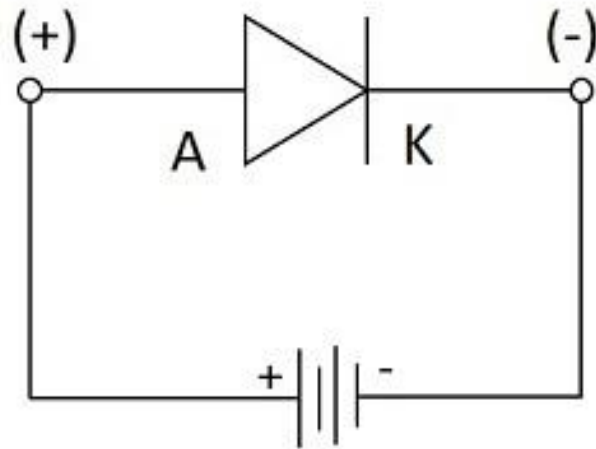
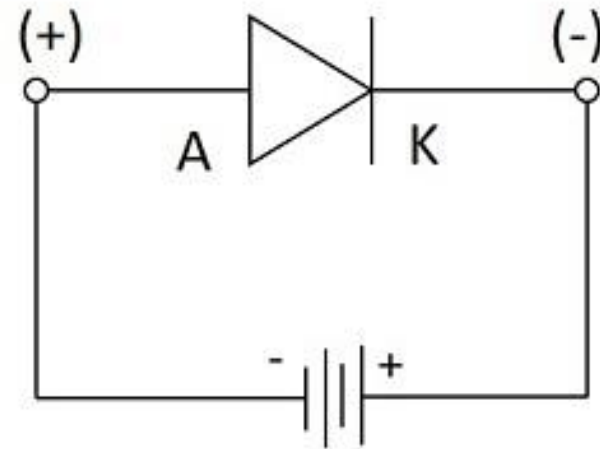


Diode Applications

Forward and reverse bias



Forward biased Connection



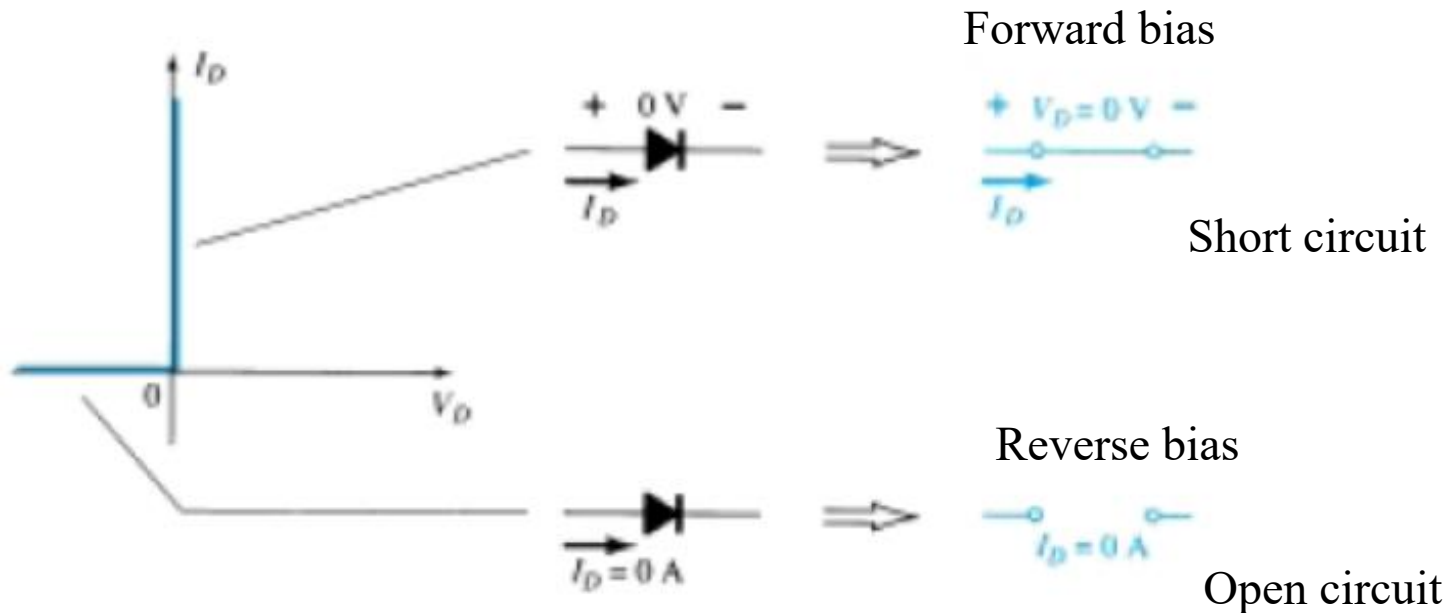
Reverse biased Connection

Ideal diode model

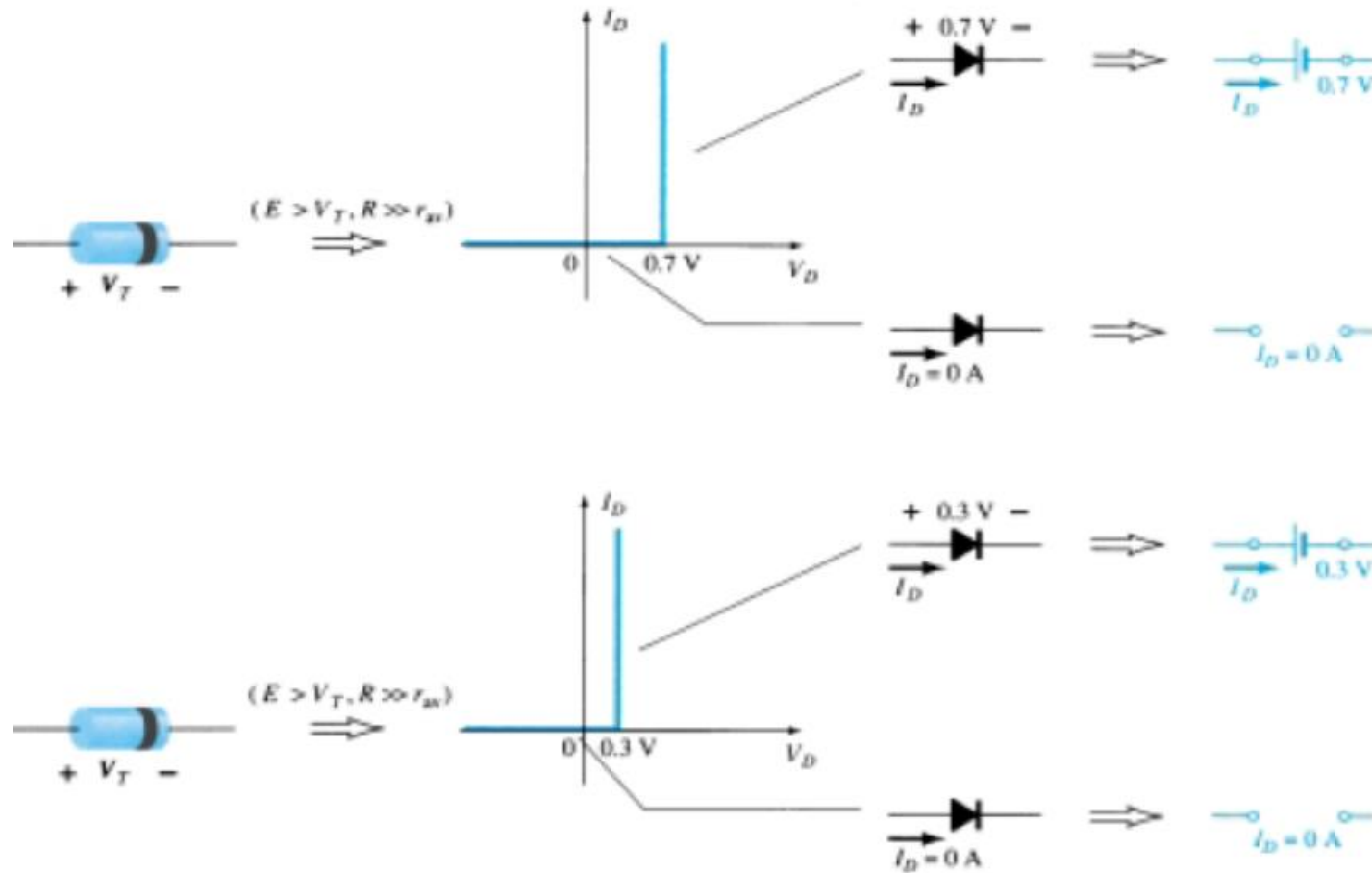
- Short circuit when forward biased
- Open circuit when reversed biased



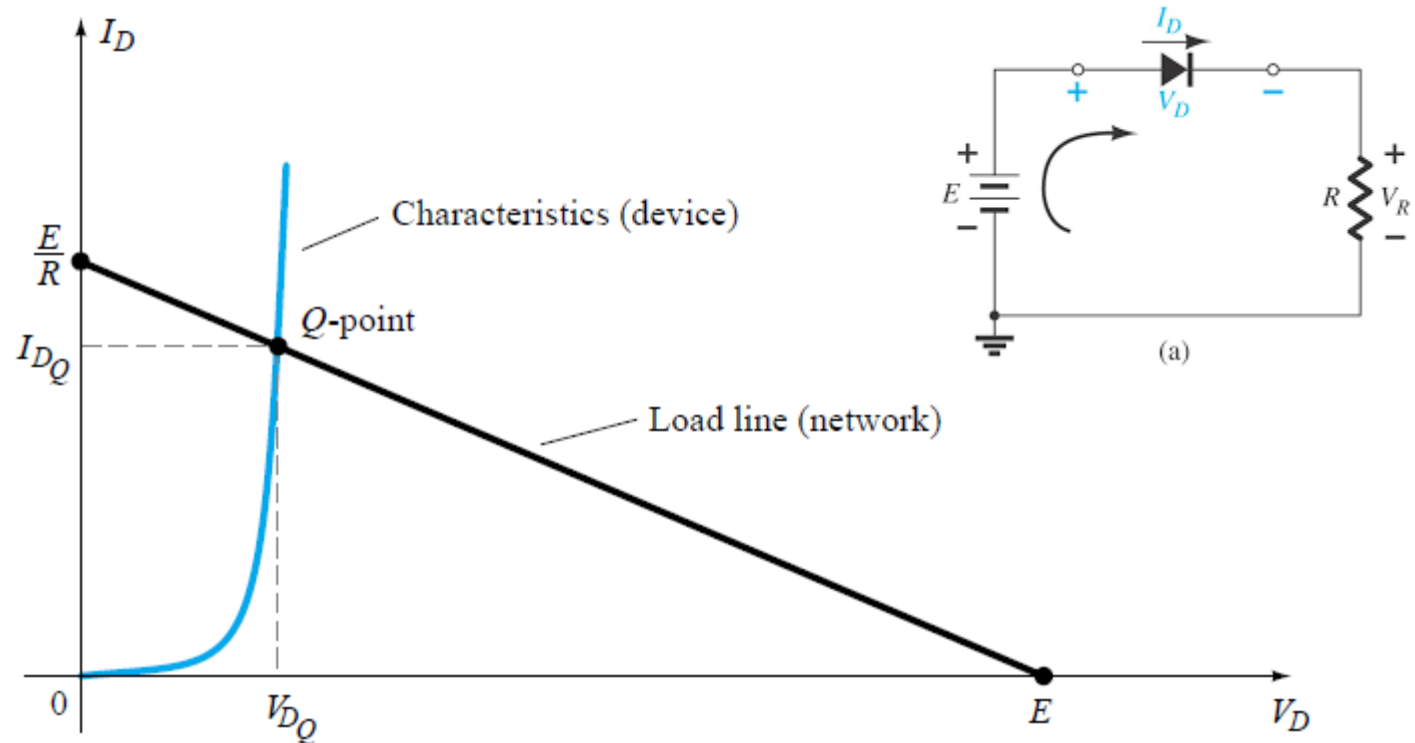
Diode and diode symbol



Practical diode simplified model



Load-Line Analysis



Drawing the load line and finding the point of operation.

$$I_D = \frac{E}{R} \Big|_{V_D=0 \text{ V}}$$

$$V_D = E \Big|_{I_D=0 \text{ A}}$$

Series diode circuit

- Determine the state of the diode first.
- Si diodes need 0.7 V and Ge diodes need 0.3 V to turn on.
- Replace the diode with a model.
- Connect a series resistance in any diode circuit to limit the current.

Analysis:

$$V_D = V_T$$

$$V_R = E - V_T$$

$$I_D = I_R = \frac{V_R}{R}$$

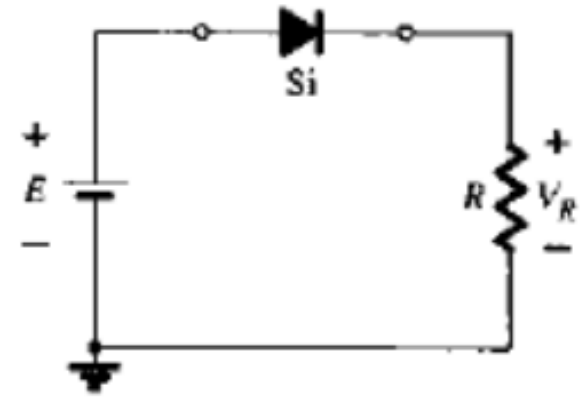


Figure 2.10 Series diode configuration.

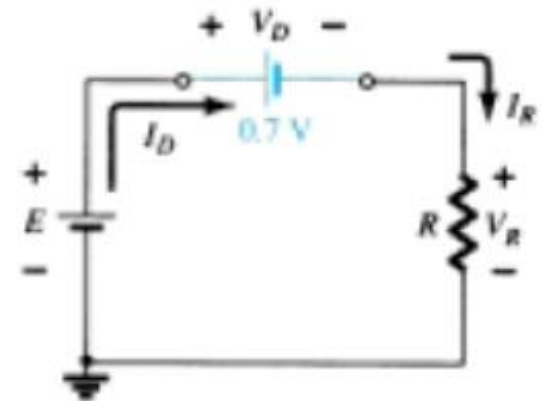


Figure 2.12 Substituting the equivalent model for the "on" diode of Fig. 2.10.

Reversed biased diode

- In this circuit of Fig. 2.13 the diode is reversed biased.
- So, in fig. 2.15, the diode is replaced by its reversed biased model, a open circuit.

Analysis:

$$V_R = I_R R = I_D R = (0 \text{ A}) R = 0 \text{ V}$$

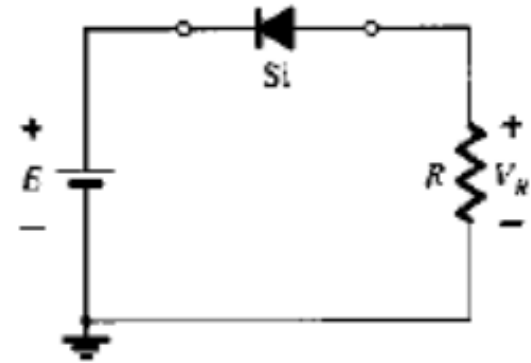


Figure 2.13 Reversing the diode of Fig. 2.10.

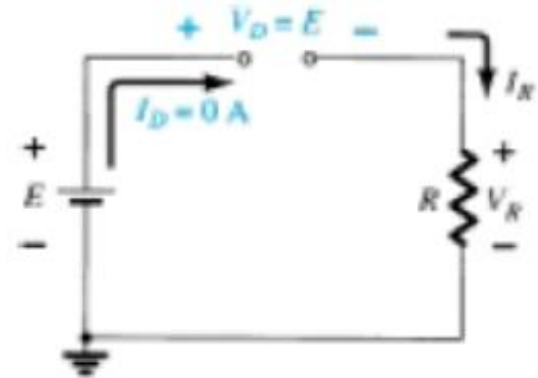


Figure 2.15 Substituting the equivalent model for the "off" diode of Figure 2.13.

Si and Ge diodes in series

EXAMPLE 2.9

Determine V_o and I_D for the series circuit of Fig. 2.21.

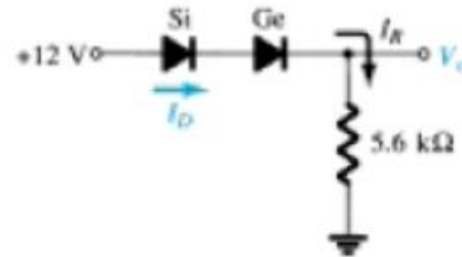


Figure 2.21 Circuit for Example 2.9.

Solution

$$V_o = E - V_{T_1} - V_{T_2} = 12 \text{ V} - 0.7 \text{ V} - 0.3 \text{ V} = 11 \text{ V}$$

and

$$I_D = I_R = \frac{V_R}{R} = \frac{V_o}{R} = \frac{11 \text{ V}}{5.6 \text{ k}\Omega} \cong 1.96 \text{ mA}$$

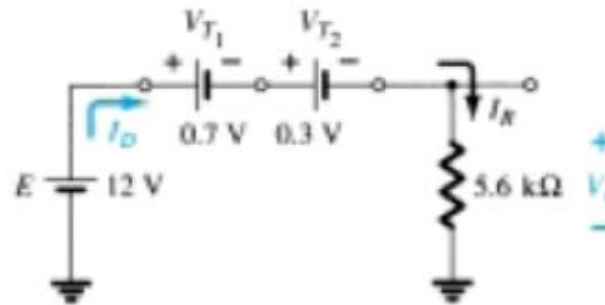


Figure 2.22 Determining the unknown quantities for Example 2.9.

Back to back diode connection

EXAMPLE 2.10

Determine I_D , V_{D_2} , and V_o for the circuit of Fig. 2.23.

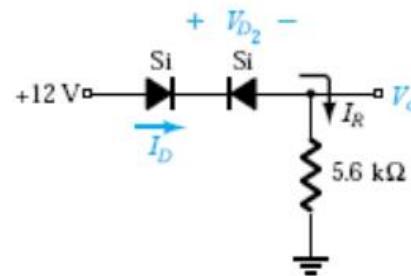


Figure 2.23 Circuit for Example 2.10.

Solution

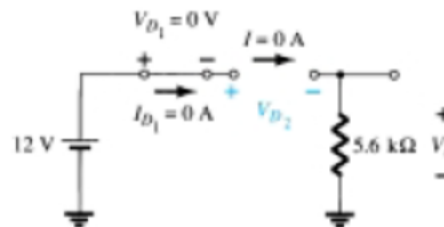


Figure 2.26 Determining the unknown quantities for the circuit of Example 2.10.

$$V_o = I_R R = I_D R = (0 \text{ A}) R = 0 \text{ V}$$

and

$$V_{D_2} = V_{\text{open circuit}} = E = 12 \text{ V}$$

Applying Kirchhoff's voltage law in a clockwise direction gives us

$$E - V_{D_1} - V_{D_2} - V_o = 0$$

and

$$\begin{aligned} V_{D_2} &= E - V_{D_1} - V_o = 12 \text{ V} - 0 - 0 \\ &= 12 \text{ V} \end{aligned}$$

with

$$V_o = 0 \text{ V}$$

Parallel diodes

EXAMPLE 2.12

Determine V_o , I_1 , I_{D1} , and I_{D2} for the parallel diode configuration of Fig. 2.30.

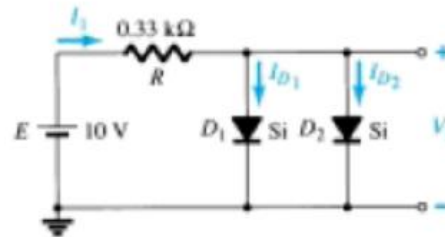


Figure 2.30 Network for Example 2.12.

Solution

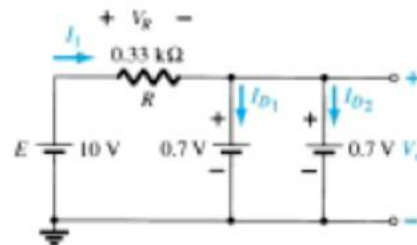


Figure 2.31 Determining the unknown quantities for the network of Example 2.12.

The current

$$I_1 = \frac{V_R}{R} = \frac{E - V_D}{R} = \frac{10 \text{ V} - 0.7 \text{ V}}{0.33 \text{ k}\Omega} = 28.18 \text{ mA}$$

Assuming diodes of similar characteristics, we have

$$I_{D1} = I_{D2} = \frac{I_1}{2} = \frac{28.18 \text{ mA}}{2} = 14.09 \text{ mA}$$

Parallel reverse diodes

Determine the current I for the network of Fig. 2.32.

EXAMPLE 2.13

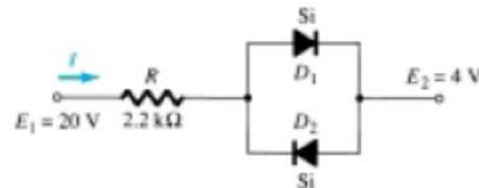


Figure 2.32 Network for Example 2.13.

Solution

Redrawing the network as shown in Fig. 2.33 reveals that the resulting current direction is such as to turn on diode D_1 and turn off diode D_2 . The resulting current I is then

$$I = \frac{E_1 - E_2 - V_D}{R} = \frac{20 \text{ V} - 4 \text{ V} - 0.7 \text{ V}}{2.2 \text{ k}\Omega} \cong 6.95 \text{ mA}$$

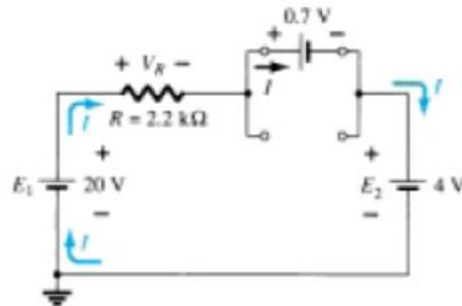


Figure 2.33 Determining the unknown quantities for the network of Example 2.13.

Si and Ge diodes in parallel

EXAMPLE 2.14

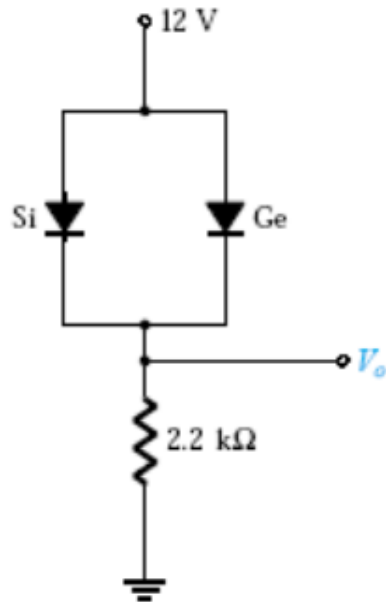


Figure 2.34 Network for Example 2.14.

$$V_o = 12 \text{ V} - 0.3 \text{ V} = 11.7 \text{ V}$$

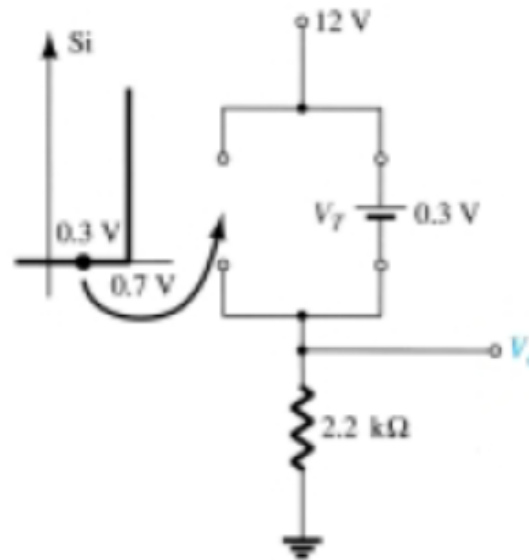


Figure 2.35 Determining V_o for the network of Fig. 2.34.

Series-parallel diodes

EXAMPLE 2.15

Determine the currents I_1 , I_2 , and I_{D_2} for the network of Fig. 2.36.

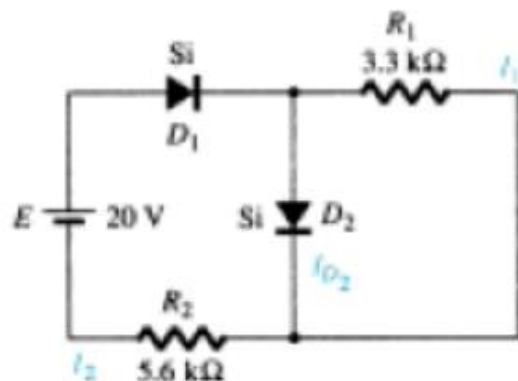


Figure 2.36 Network for Example 2.15.

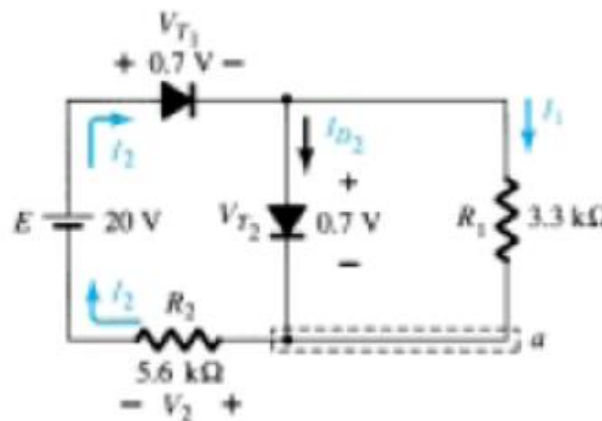


Figure 2.37 Determining the unknown quantities for Example 2.15.

$$I_1 = \frac{V_{T_2}}{R_1} = \frac{0.7 \text{ V}}{3.3 \text{ k}\Omega} = 0.212 \text{ mA}$$

Applying Kirchhoff's voltage law around the indicated loop in the clockwise direction yields

$$-V_2 + E - V_{T_1} - V_{T_2} = 0$$

and
$$V_2 = E - V_{T_1} - V_{T_2} = 20 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V} = 18.6 \text{ V}$$

with
$$I_2 = \frac{V_2}{R_2} = \frac{18.6 \text{ V}}{5.6 \text{ k}\Omega} = 3.32 \text{ mA}$$

At the bottom node (a),

$$I_{D_2} + I_1 = I_2$$

and
$$I_{D_2} = I_2 - I_1 = 3.32 \text{ mA} - 0.212 \text{ mA} = 3.108 \text{ mA}$$

Application of dc diode circuits

- To drive LEDs in power indicators circuits
- LED Lighting
- Logic circuits, etc

Points to keep in mind

- Before using a diode in a circuit, you must know at least the following:

Its forward current I_F ,

Zener potential V_Z ,

Maximum power dissipation ($P_{D,max}$),

and operating temperature range.

Home work

- Solve problems at the end of Chapter 2
- Problem no. 6 to 11 and 13.

THANKS